

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Double-Blind, Placebo-Controlled Study of Talazoparib with Enzalutamide in Metastatic Castration-Resistant Prostate Cancer **Short Title/Acronym:** TALAPRO-2 / Talazoparib + Enzalutamide vs. Enzalutamide Monotherapy in mCRPC **Protocol Number:** 2017-003295-31 / STUDY00144652 **NCT Number:** NCT03395197 **Posting ID:** 881148931849906287 **Sponsor/Company Name:** Pfizer **Investigational Product:** Talazoparib (Trade name: Talzenna; ATC: L01XK04; PARP inhibitor) in combination with Enzalutamide **Phase of Development:** Phase 3 **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective (Part 1): To confirm the starting dose of talazoparib in combination with enzalutamide through assessment of target safety events and pharmacokinetics (PK). **Primary Objective (Part 2):** To evaluate the radiographic progression-free survival (rPFS) in patients with metastatic castration-resistant prostate cancer (mCRPC) treated with talazoparib plus enzalutamide versus placebo plus enzalutamide. **Secondary Objectives:** To evaluate Overall Survival (OS), Objective Response Rate (ORR), time to confirmed Prostate-Specific Antigen (PSA) progression, safety, and patient-reported outcomes in both all-comers and homologous recombination repair (HRR) / DNA damage response (DDR) gene-mutated cohorts.

2.2 Background & Rationale To address the need for improved first-line therapies in mCRPC. PARP inhibitors in combination with androgen receptor-signaling inhibitors (like enzalutamide) have demonstrated synergistic potential in reducing tumor size and slowing tumor growth in mCRPC patients. This study aims to generate clinical evidence on whether this combination improves survival outcomes in molecularly unselected patients (all-comers) and in those with DDR / HRR gene alterations.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Trial Structure:**
- **Part 1:** Open-label, non-randomized, safety and PK run-in study to confirm the starting dose at select sites. - **Part 2:** Randomized (1:1), double-blind, placebo-controlled, multinational

study. **Control Type:** Placebo-controlled (for Part 2) **Blinding:** Double-blind (for Part 2)

3.2 Planned Enrollment & Duration Target **SN\$:** 1,054 patients
Number of Sites: Multinational (across 27 countries) **Estimated Study Start:** 18-Dec-2017 **Estimated Primary Completion:** 03-Oct-2022 (Actual)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically or cytologically confirmed adenocarcinoma of the prostate without small cell or signet cell features. 2. Asymptomatic or mildly symptomatic metastatic castration-resistant prostate cancer (mCRPC) (score on BPI-SF Question #3 must be < 4). 3. Surgically or medically castrated, with serum testosterone ≤ 50 ng/dL (≤ 1.73 nmol/L) at screening. 4. Metastatic disease in bone documented on bone scan or in soft tissue documented on CT/MRI scan. 5. Progressive disease at study entry defined by ≥ 1 of 3 criteria: PSA progression (minimum of 2 rising values), soft tissue progression per RECIST 1.1, or bone progression per PCWG3 (≥ 2 new lesions). 6. Eastern Cooperative Oncology Group (ECOG) performance status ≤ 1 . 7. Life expectancy ≥ 12 months as assessed by the investigator. 8. Able to swallow the study drug and have no known intolerance to study drugs. 9. Must consent to saliva sample collection for DDR/germline comparator (optional in Part 1) and agree to stringent highly effective contraception/condom use and not to donate sperm through 4 months after the last dose.

4.2 Exclusion Criteria 1. Any prior systemic cancer treatment initiated in the non-metastatic CRPC and mCRPC disease state (except prior ADT/first-generation antiandrogens). 2. Patients whose only evidence of metastasis is adenopathy below the aortic bifurcation. 3. Prior treatment with second-generation androgen receptor inhibitors (enzalutamide, apalutamide, darolutamide), a PARP inhibitor, cyclophosphamide, or mitoxantrone for prostate cancer. 4. Prior treatment with platinum-based chemotherapy within 6 months prior to Day 1 (Part 1) or randomization (Part 2), or disease progression on platinum-based therapy within 6 months. 5. Prior treatment with opioids for prostate cancer-related pain within 28 days prior to Day 1/randomization. 6. Current use of potent P-gp inhibitors within 7 days. 7. Significant renal dysfunction (eGFR < 30 mL/min/1.73 m²; for Part 1 only, moderate impairment eGFR 30-59 mL/min/1.73 m² is also excluded). 8. Significant hepatic dysfunction (Bilirubin $> 1.5 \times$ ULN, AST or ALT $> 2.5 \times$ ULN, Albumin < 2.8 g/dL). 9. Hematologic inadequacies: ANC $< 1500/\mu$ L, platelets $< 100,000/\mu$ L, or hemoglobin < 9 g/dL. 10. Known or suspected brain metastasis, active leptomeningeal disease, symptomatic/impending spinal cord compression, or cauda equina

syndrome. 11. History of seizure, loss of consciousness, or transient ischemic attack (TIA) within 12 months.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency (Part 2):

Talazoparib 0.5 mg once daily + Enzalutamide 160 mg once daily vs. Placebo once daily + Enzalutamide 160 mg once daily. **Route of Administration:** Oral **Duration of Treatment:** Until disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Ongoing androgen deprivation therapy (ADT) with a GnRH agonist or antagonist; ongoing bisphosphonate or denosumab use prior to Day 1/ randomization is allowed but not mandatory. **Prohibited:** Concurrent systemic anticancer therapy, other PARP inhibitors, or second-generation antiandrogens not specified in the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to 0	Screening: Informed Consent, Medical History, Tumor tissue/saliva collection for HRR/germline status, Imaging (CT/MRI/Bone Scan), Labs
2	Day 1	Baseline: Randomization (Part 2), First Dose, Safety Labs
3-10	Every 4-8 weeks	Interim: Safety and Efficacy Assessments, Imaging, PSA tests, PK analysis
11	+28 Days post-last dose	End of Study: Safety Follow-up, Final Vital Signs, Exit Interview, Long-term OS Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints - Part 1 (Safety Run-in): - Number of Participants With Treatment-Emergent Adverse Events (TEAEs) Occurring Within the First 66 Days of Dosing. - Number of Participants With All-Causality Clustered Treatment-Emergent Cytopenias (Anemia, Thrombocytopenia, Neutropenia, Leukopenia) by Preferred Term (PT) and Max CTCAE Grade Occurring Within the First 66 Days of Dosing. - Number of Participants With All-Causality TEAEs During the Overall Period of Part 1. - **Part 2 (Efficacy):** - Radiographic progression-free survival (rPFS) defined as time from randomization to objective radiographic progression or death, measured via Blinded Independent Central Review (BICR) per RECIST 1.1 (soft tissue) and PCWG3 criteria (bone).

7.2 Secondary & Exploratory Endpoints - Overall Survival (OS) - Objective Response Rate (ORR) - Time to confirmed Prostate-Specific Antigen (PSA) progression.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection focusing on known PARPi and enzalutamide toxicities (e.g., cytopenias including anemia and neutropenia, fatigue, fractures). Graded via CTCAE version 4. **Serious Adverse Events (SAEs):** Standard reporting within 24 hours of site awareness for events resulting in death, hospitalization, life-threatening experience, or significant disability. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee to review unblinded safety and efficacy data throughout the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Evaluated in two co-primary cohorts for Part 2: Cohort 1 (all-comers) and Cohort 2 (patients with HRR / DDR gene alterations). Safety Set includes all patients who received ≥ 1 dose of study drug. **Sample Size Justification:** Powered to detect a statistically significant improvement in rPFS for the talazoparib + enzalutamide arm vs. placebo + enzalutamide arm. **Handling of Missing Data:** Standard survival analysis censoring rules applied for PFS and OS drop-outs.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring to ensure protocol adherence, subject safety, and data integrity. **Audit Trail:** Electronic Data Capture (EDC) system implementation with full audit trails and data clarification mechanisms.

11. Study Identifiers & Governance

Primary ID: STUDY00144652 / 2017-003295-31 **Posting ID:** 881148931849906287 **Registry Number:** NCT03395197 **Sponsor/Lead Organization:** Pfizer **Study Title:** Talazoparib + Enzalutamide vs. Enzalutamide Monotherapy in mCRPC **Official Regulatory Title:** A Phase 3, Randomized, Double-Blind, Placebo-Controlled Study of Talazoparib with Enzalutamide in Metastatic Castration-Resistant Prostate Cancer

12. Clinical Framework (Oncology / Prostate Cancer Specific)

Primary Condition: Metastatic Castration-Resistant Prostate Cancer (mCRPC) (Preferred UMLS Name: Treatment of Metastatic Castration-Resistant Prostate Cancer) **Semantic Entity:**

Intermediate Atypical Prostate Carcinoma / Neoplastic Process
Diagnosis Threshold: Histologically/cytologically confirmed adenocarcinoma of the prostate, castrate levels of testosterone (≤ 50 ng/dL), and evidence of metastasis. **Medical Methodology:** First-line systemic combination therapy for mCRPC
Optimization Target: Delay of radiographic progression and improvement of overall survival.

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway for mCRPC
Education Protocol (HCP): Protocol training, RECIST 1.1 and PCWG3 criteria evaluation training
Education Protocol (Patient): Study requirements, AE reporting instructions, study drug dosing compliance
Quality Audit Mechanism: Centralized imaging review (BICR) to confirm rPFS and objective response independently.

14. Observation & Follow-Up Schedule

Total Duration: Estimated 3-5 years (until end of overall survival follow-up)
Onsite Visit Cadence: Every 4-8 weeks during the active treatment phase
Remote Monitoring Frequency: Periodic follow-ups for long-term survival after treatment discontinuation
Checklist Review Interval: Routine reviews for pill count accountability and AE evaluation

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				